

Assessment #1: Industry Trends & Future Outlook Assessment

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Assessment #1 - Current State of the Sabermetrics Industry

1. What's Happening Right Now?

The rapid integration of the Automated Ball-Strike System (ABS) is eliminating unnecessary human error by umpires. Sabermetrics contributed to this implementation by suggesting universal adjustment based on player physical profiles (Ahn, 2026). Moreover, rather than evaluating players solely on surface outcomes like hits or strikeouts, front offices use optical tracking to measure geometric contact depth, swing intent, and subtle aerodynamic pitch movement (Nappert, 2026).

2. What's Disrupting the Field?

Researchers are adapting Large Language Model (LLM) architectures to train millions of sequential pitch-by-pitch Statcast tokens, allowing teams to simulate batter decision-making and pitch dynamics with high precision (Ahn, 2026). Modern analytics successfully optimized pitch velocity and spin rates, but this pushed human joints and muscles to physical limits that cannot be bypassed, causing high rates of elbow and shoulder injuries from extremity; essentially, all top-performing pitchers will eventually get injuries and following surgeries in their optimization process. Therefore, the current debate centers on whether the optimization technology should be developed further in return of the increasing rate of injuries if not using biomechanical modeling to reduce joint stress without sacrificing pitch quality (Fleisig, 2026).

3. Old Ways vs. New Ways

- Past Approach: Heavy reliance on traditional box-score metrics (Batting Average, RBI, ERA) and subjective visual scouting did not successfully eliminate random variance (Dillon).
- Modern Approach: trajectory tracking, kinetic chain modeling (measuring ground reaction forces and arm-slot variance to analyze explosiveness efficiency), and predictive metrics like expected weighted

On-base Average (xwOBA) and Vertical Approach Angle (VAA), successfully establishing a foundation for player evaluation with minimal noise. The emergence of high-performing AI facilitated meticulous and time-demanding analysis; the issue is shifting towards the collection of enough data (high-frame videos, spreadsheets, etc.) for inputting AI to yield accurate result (Kamino).

4. Looking Ahead & Keeping Your Research Fresh

I expect analytics to move toward real-time biomechanical diagnostics. Within a few seasons, dugouts will likely use AI alerts that detect microscopic movement shifts mid-game to predict pitcher fatigue before injuries or performance drops occur. This trend shifts baseball analytics from descriptive evaluation (measuring past value) to predictive movement analysis (optimizing human motion and physical decision-making) (Ahn).

5. Industry Publications & Trend Tracking

- Baseball Savant (baseballsavant.mlb.com)

I will bookmark their Statcast leaderboards to monitor shifting trends in bat tracking, pitch movement, and tactics trends throughout the MLB season.

- FanGraphs (fangraphs.com)

I will subscribe to their daily newsletter to read current community research and in-depth quantitative player scouting reports.

- Baseball Prospectus (baseballprospectus.com)

I will follow their social media accounts and technical research articles to track updates to projection models like PECOTA, Repsodo, etc.

- Society for American Baseball Research (sabr.org)

I will review annual SABR Analytics Conference presentations and subscribe to the SABR Baseball Research Journal for peer-reviewed academic studies.

Place APA citation(s) at the end (in order of author). Don't forget in-text citations within your assessment!

- Ahn, Y. J., Du, Y., Zhang, Z., & Kang, H. (2026). *Neural Sabermetrics with World Model: Play-by-play Predictive Modeling with Large Language Model*. arXiv.Org. <https://arxiv.org/abs/2602.07030>
- Dillon, M., Frappa, N., Mastroianni, M. A., Ablove, R. A., & Nicholson, K. F. (2026). Direct arm angle versus release-based proxies in Major League Baseball pitchers: a large-scale validation study. *Sports Biomechanics*, 1–10. <https://doi.org/10.1080/14763141.2026.2638934>
- Fleisig, G. S. (2026). Baseball has an elbow injury problem: biomechanics may have the solution. *Sports Biomechanics*, 1–19. <https://doi.org/10.1080/14763141.2026.2689508>
- Nappert P, Plante M, Bamber M (2026), "Playing ball: evaluation, valuation, and accountants in Major League Baseball". *Qualitative Research in Accounting & Management*, Vol. 23 No. 1 pp. 90–117, doi: <https://doi.org/10.1108/QRAM-11-2024-0240>
- Waki Kamino, Wen-Yi, A. W., Sabanovic, S., & Jung, M. (2026). Beyond Accuracy: Rethinking the Value of AI in Decision-Making Through Baseball's Automated Ball-Strike (ABS) System. Hicss 2026. https://www.researchgate.net/profile/Waki-Kamino/publication/398934388_Beyond_Accuracy_Rethinking_the_Value_of_AI_in_Decision-Making_Through_Baseball